Roll Number												

Narula Institute of Technology
An Autonomous Institute under MAKAUT
2024
END SEMESTER EXAMINATION - EVEN 2024
PH(CS)201 - ENGINEERING PHYSICS

TIME ALLOTTED: 3Hours **FULL MARKS: 70**

Instructions to the candidate:

Figures to the right indicate full marks.
Draw neat sketches and diagram wherever is necessary.
Candidates are required to give their answers in their own words as far as practicable

Group A
(Multiple Choice Type Questions)
Answer any ten from the following, choosing the correct alternative of each question: 10×1=10

1.i) The dimension of the Compton wavelength will be (1) CO2 BL3

a) [ML]
b) [L]
c) [LT]
d) [T]

1.ii) For p-type semiconductor, impurity injected to intrinsic semi conductor is (1) CO1 BL2

a) boron
b) aluminium
c) gallium
d) All of the above

1.iii) For a quantum wave particle, E = (1) CO5 BL1

$\hbar$ k

a)

b) $\hbar$ ω
c) $\hbar\omega/2$
d) $\hbar k/2$
e) None of the above

1.iv) Size confinement is maximum in (1) CO2 BL2

a) Bulk material
b) Quantum dot
c) Quantum wire

d) Quantum well

1.v) The operation of the fibre optic cable is based on the principle of (1) CO1 BL3
a) diffraction
b) reflection
c) dispersion
d) total internal reflection

1.vi) What are the Miller indices of the plane having intercepts 2a, 3b, 1c ? (1) CO3 BL3
a) (326)
b) (236)
c) (263)
d) (216)

1.vii) Strongest material of the world is (1) CO1 BL1
a) Carbon steel
b) Graphene
c) Steel
d) ALNICO

1.viii) What is the frequency of optical signal ? (1) CO1 BL1
a) 10^5 Hz
b) 10^8 Hz
c) 10^{15} Hz
d) 10^{11} Hz

1.ix) Life time of ordinary excited states is (1) CO1 BL2
a) $\mathbf{10}^{-13}$S
b) $\mathbf{10}^{-15}$S
c) 100 S
d) $\mathbf{10}^{-8}$ S

1.x) Holography is a (1) CO1 BL4
a) lens-less imaging process
b) lens-less 2-D imaging process
c) lens-less 3-D imaging process
d) 3-D imaging process

1.xi) In USB storage COFEE represents (1) CO4 BL5
a) Computer On Forensic Evidence Extractor
b) Computer Online Forensic Evidence Extractor
c) Computer Online Foreign Evidence Extractor
d) Computer Online Forensic Effect Extractor

1.xii) The liquid crystals used in the LCD are (1) CO4 BL1
a) Twisted Nemantic
b) Tallest Nemantic
c) Twisted Nemantis
d) Tallest Nemantis

Group B
(Short Answer Type Questions)
(Answer any three of the following) 3x5=15

2. a) Write down the expression for phase velocity and group velocity explaining all the terms. Write down the relation between them. b) A ball of mass 1kg is moving with a velocity of 10m/s. Find out the de-Broglie wavelength of the ball.	(5)	CO5	BL3
3. Describe the working principle of He-Ne laser with the help of energy level diagram of it.	(5)	CO1	BL2
4. a) Calculate the wavelength associated with an electron and a photon carrying energy 100eV. b) Write down the expression for the Compton shift of an electron. At which angle the shift will be maximum? Calculate the Compton wavelength for an electron.	(5)	CO5	BL3
5. Derive the expression of Einstein A and B coefficients.	(5)	CO1	BL3
6a. Can you measure the correct value of the position and momentum of an object simultaneously -Justify.	(2)	CO5	BL5
6b. Is it possible to liberate an electron from a metal surface having a work function of 4.8 eV with an incident radiation of wavelength 5000 A^0?	(3)	CO3	BL5

Group C
(Long Answer Type Questions)
(Answer any three of the following) 3x15=45

7. a) A proton and electron have the same wavelength. Which one has higher energy? b) The work function of a metal surface is 5.6 eV. Photo electrons of 6 eV are ejected. Find out the wavelength of radiation used. c) In a crystal system, the lattice parameters a = b = 2.5 Å, c = 1.8 Å, find the interplaner spacing. d) Describe the construction and working principle of Ruby Laser with energy level diagrams.	(15)	CO5	BL3
8. a) What is the Compton Effect? What is Planck's quantum hypothesis? b) Discuss the intrinsic and extrinsic semiconductors. Give examples also. c) Describe the working principle of Liquid crystal display (LCD), LED, OLED. Discuss the several applications of LCD.	(15)	CO4	BL2
9. a) The spacing between the principal planes of NaCl crystal is 2.82 Å. It is found that the first-order reflection of monochromatic X-rays occurs at an angle 10 degrees. What is the wavelength of X-rays? b) Calculate the wavelength of the laser emitted from an extrinsic semiconductor laser if the band gap is 0.02eV. To which region of the	(15)	CO5	BL3

EM spectrum does it belong?
c) The refractive indices of core and cladding are 1.50 and 1.48 respectively in an optical fibre. Find the numerical aperture and acceptance angle.

d) A radiation of wavelength 300 nm is incident on a silver surface. Will photoelectrons be observed? [work function of silver = 4.7 eV]

e) When the light of wavelength 2200Å falls on Cu, photoelectrons are emitted from it. Find (i) the threshold wavelength and (ii) the stopping potential. Given: the work function for Cu is 4.65 eV.

10. a) Explain the terms i) spontaneous emission, ii) stimulated emission, and iii) stimulated absorption. Distinguish between spontaneous and stimulated emission of radiation. (15) CO2 BL3
b) State the differences between Step-Index and Graded-Index optical fibre. Explain the process of hologram recording, including the steps involved in capturing and storing the interference pattern.

c) “One nanometre is a magical point on the dimension scale.” Explain. Highlight the properties of carbon nanotubes. Give some present and future applications of nanomaterials.

d) The wavelength of an electron and a photon is 1.00 nm. Determine

(i) their momenta,

(ii) photon energy,

(iii) electron kinetic energy.

11a. How Liquid Crystal Display work ? (5) CO1 BL4

11b. Obtain the Packing fraction for BCC crystal . (5) CO3 BL3

11c. Find the energy difference between two energy levels of an atomic system if the transition between these levels gives a photon of wavelength 694.3 nm. Also calculate the no. of photons emitted to give a power output of 3 mw. (5) CO1 BL3

6/18/2024 11:42:55 AM